

MEDICAL INSURANCE DATA ANALYSIS PROJECT

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SETTING WORKING DIRECTORY

```
setwd("C:/Users/HomePC/Desktop/e-commerce project")
```

DATA IMPORTATION

```
insurance_data<-read.csv("insurance_data.csv")
head(insurance_data)
```

	age	sex	bmi	children	smoker	region	charges
1	19	female	27.900	0	yes	southwest	16884.924
2	18	male	33.770	1	no	southeast	1725.552
3	28	male	33.000	3	no	southeast	4449.462
4	33	male	22.705	0	no	northwest	21984.471
5	32	male	28.880	0	no	northwest	3866.855
6	31	female	25.740	0	no	southeast	3756.622

structure

```
str(insurance_data)
```

```
'data.frame': 1338 obs. of 7 variables:
 $ age      : int  19 18 28 33 32 31 46 37 37 60 ...
 $ sex      : chr  "female" "male" "male" "male" ...
 $ bmi      : num  27.9 33.8 33 22.7 28.9 ...
 $ children: int   0 1 3 0 0 0 1 3 2 0 ...
 $ smoker   : chr   "yes" "no" "no" "no" ...
 $ region   : chr   "southwest" "southeast" "southeast" "northwest" ...
 $ charges  : num  16885 1726 4449 21984 3867 ...
```

DATA CLEANING

missing values

```
missing_value<-sum(is.na(insurance_data))
print(missing_value)
```

```
[1] 0
```

removing duplicates

```
duplicated_values<-sum(duplicated(insurance_data))
print(duplicated_values)
```

```
[1] 1
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
insurance_data <- distinct(insurance_data)
```

```
duplicated_values<-sum(duplicated(insurance_data))  
print(duplicated_values)
```

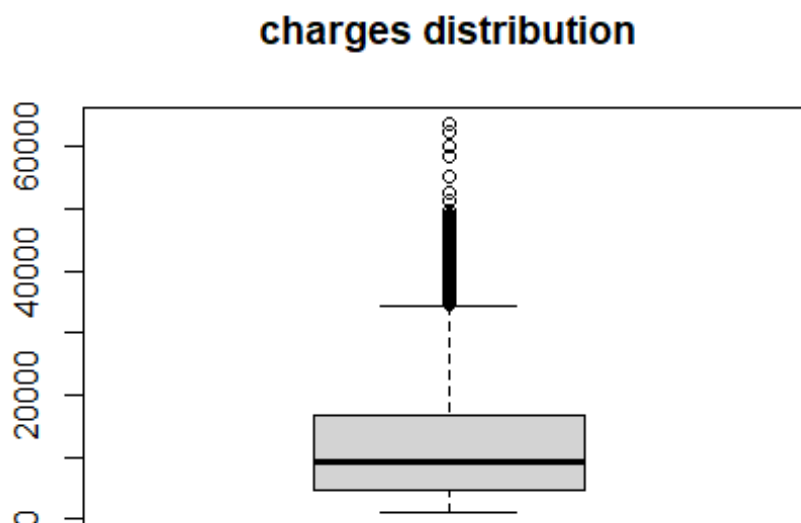
```
[1] 0
```

```
unique(insurance_data$sex)
```

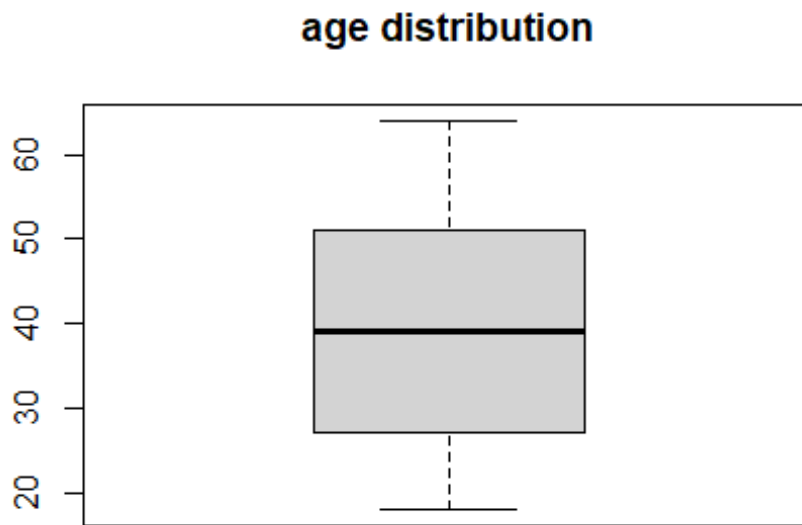
```
[1] "female" "male"
```

detecting outlier

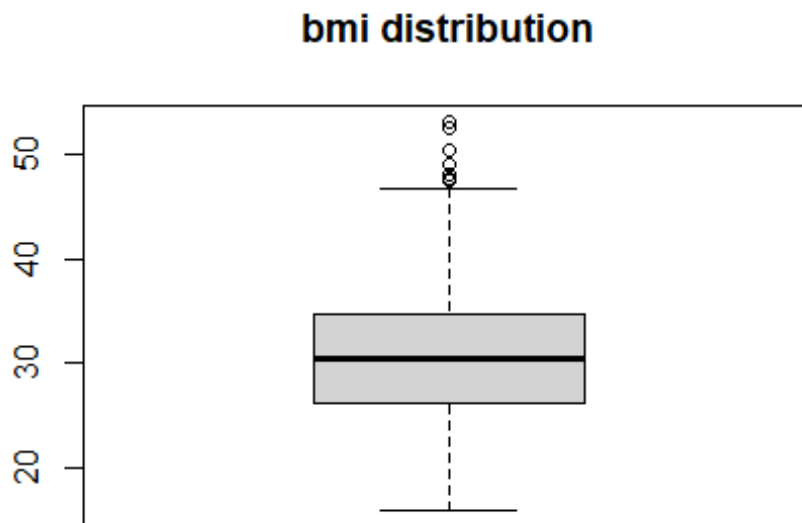
```
boxplot(insurance_data$charges,main="charges distribution")
```



```
boxplot(insurance_data$age, main = "age distribution")
```



```
boxplot(insurance_data$bmi, main="bmi distribution")
```



treating outliers

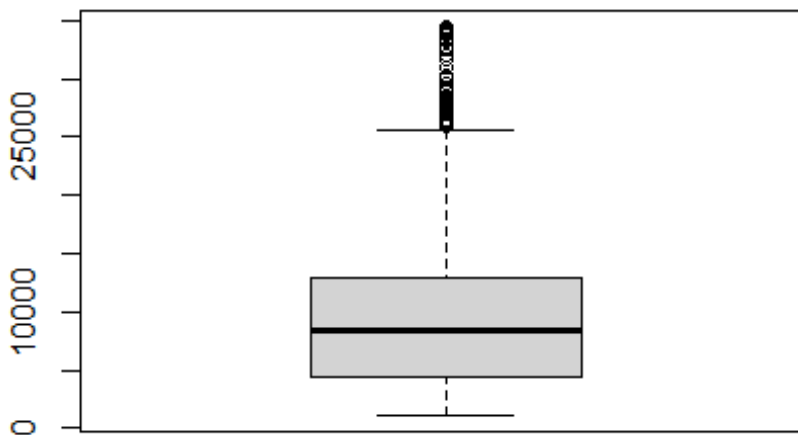
```
Q1 <- quantile(insurance_data$charges, 0.25)
Q3 <- quantile(insurance_data$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

d_cleaned <- insurance_data[insurance_data$charges >= lower_bound & insurance_data$charges <= upper_bound,]

Q1 <- quantile(d_cleaned$bmi, 0.25, na.rm = TRUE)
Q3 <- quantile(d_cleaned$bmi, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

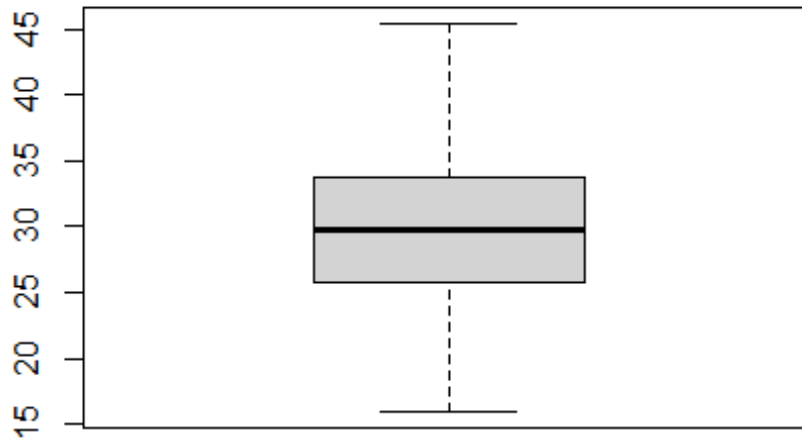
d_cleaned <- d_cleaned[d_cleaned$bmi >= lower_bound & d_cleaned$bmi <= upper_bound,]
boxplot(d_cleaned$charges, main="charges distribution")
```

charges distribution



```
boxplot(d_cleaned$bmi, main="bmi distribution")
```

bmi distribution

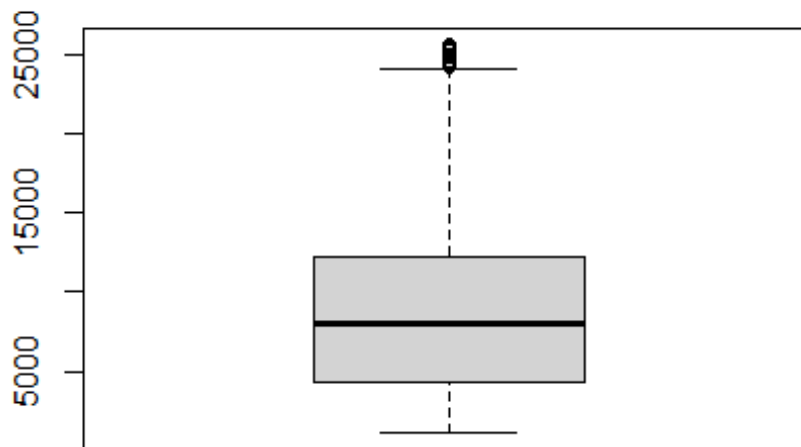


```
Q1 <- quantile(d_cleaned$charges, 0.25, na.rm = TRUE)
Q3 <- quantile(d_cleaned$charges, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

g_cleaned <- d_cleaned[d_cleaned$charges >= lower_bound & d_cleaned$charges <= upper_bound,]

boxplot(g_cleaned$charges, main="charges distribution")
```

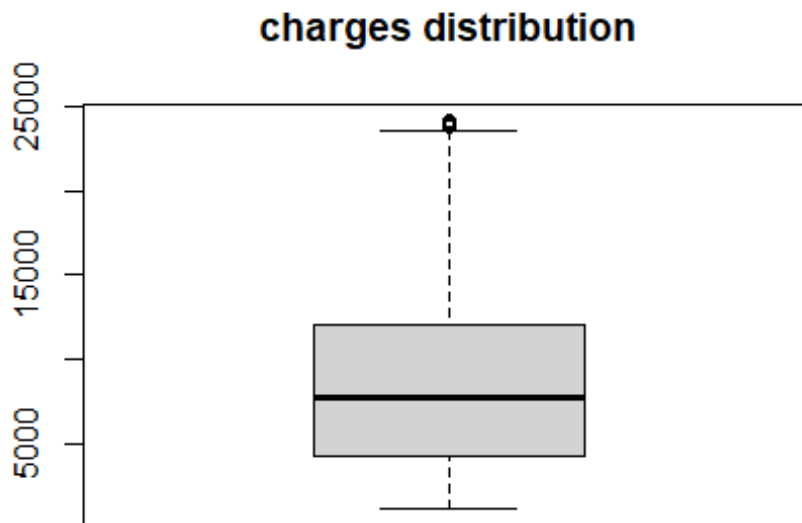
charges distribution



```
Q1 <- quantile(g_cleaned$charges, 0.25, na.rm = TRUE)
Q3 <- quantile(g_cleaned$charges, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

g_cleaned <- g_cleaned[g_cleaned$charges >= lower_bound & g_cleaned$charges <= upper_bound,]

boxplot(g_cleaned$charges, main="charges distribution")
```

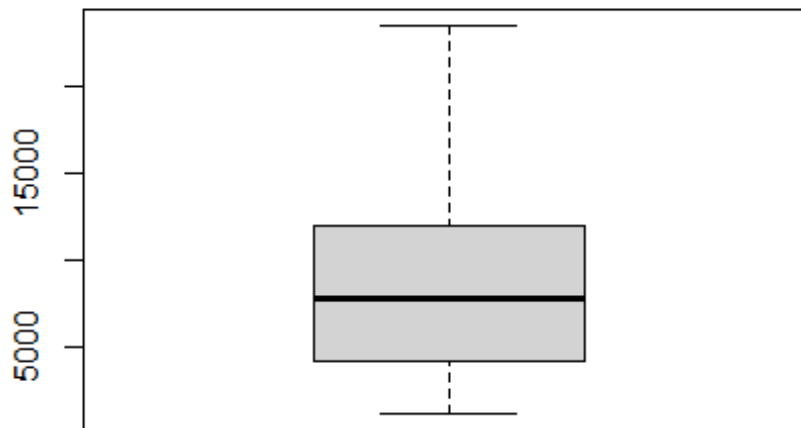


```
Q1 <- quantile(g_cleaned$charges, 0.25, na.rm = TRUE)
Q3 <- quantile(g_cleaned$charges, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

h_cleaned <- g_cleaned[g_cleaned$charges >= lower_bound & g_cleaned$charges <= upper_bound,]

boxplot(h_cleaned$charges, main="charges distribution")
```

charges distribution



```
unique(h_cleaned$children)
```

```
[1] 0 1 3 2 5 4
```

DATA MANUPULATION

```
library(dplyr)
```

```
h_cleaned <- h_cleaned %>%  
  mutate(age_group = case_when(  
    age < 20 ~ "Young",  
    age >= 20 & age <= 35 ~ "Youth",  
    age >= 36 & age <= 59 ~ "Adult",  
    age >= 60 ~ "Old"  
  ))
```

```
head(h_cleaned)
```

	age	sex	bmi	children	smoker	region	charges	age_group
1	19	female	27.900	0	yes	southwest	16884.924	Young
2	18	male	33.770	1	no	southeast	1725.552	Young
3	28	male	33.000	3	no	southeast	4449.462	Youth
4	33	male	22.705	0	no	northwest	21984.471	Youth
5	32	male	28.880	0	no	northwest	3866.855	Youth
6	31	female	25.740	0	no	southeast	3756.622	Youth

```
library(dplyr)
```

```
h_cleaned_processed1 <- h_cleaned %>%
  mutate(bmi_category = case_when(
    bmi < 18.5 ~ "Underweight",
    bmi >= 18.5 & bmi <= 24.9 ~ "Normal",
    bmi >= 25 & bmi <= 29.9 ~ "Overweight",
    bmi >= 30 ~ "Obese"
  ))
head(h_cleaned_processed1)
```

	age	sex	bmi	children	smoker	region	charges	age_group	bmi_category
1	19	female	27.900	0	yes	southwest	16884.924	Young	Overweight
2	18	male	33.770	1	no	southeast	1725.552	Young	Obese
3	28	male	33.000	3	no	southeast	4449.462	Youth	Obese
4	33	male	22.705	0	no	northwest	21984.471	Youth	Normal
5	32	male	28.880	0	no	northwest	3866.855	Youth	Overweight
6	31	female	25.740	0	no	southeast	3756.622	Youth	Overweight

```
h_cleaned_processed1 <- h_cleaned_processed1[!is.na(h_cleaned_processed1$bmi_category), ]
```

```
h_filtered <- h_cleaned_processed1 %>%
  filter(smoker == "yes", age > 30) %>%
  arrange(desc(charges))
```

```
h_filtered
```

	age	sex	bmi	children	smoker	region	charges	age_group	bmi_category
1	48	male	28.000	1	yes	southwest	23568.27	Adult	Overweight
2	47	female	26.125	1	yes	northeast	23401.31	Adult	Overweight
3	49	male	25.600	2	yes	southwest	23306.55	Adult	Overweight
4	53	female	22.880	1	yes	southeast	23244.79	Adult	Normal
5	47	female	27.830	0	yes	southeast	23065.42	Adult	Overweight
6	43	female	26.700	2	yes	southwest	22478.60	Adult	Overweight
7	56	male	19.950	0	yes	northeast	22412.65	Adult	Normal
8	40	female	28.120	1	yes	northeast	22331.57	Adult	Overweight

9	51	male	23.210	1	yes	southeast	22218.11	Adult	Norma
10	42	male	30.000	0	yes	southwest	22144.03	Adult	Obes
11	47	male	25.410	1	yes	southeast	21978.68	Adult	Overweigh
12	43	female	24.700	2	yes	northwest	21880.82	Adult	Norma
13	43	female	26.885	0	yes	northwest	21774.32	Adult	Overweigh
14	43	female	25.270	1	yes	northeast	21771.34	Adult	Overweigh
15	46	female	23.655	1	yes	northwest	21677.28	Adult	Norma
16	39	female	24.890	3	yes	northeast	21659.93	Adult	Norma
17	32	male	28.120	4	yes	northwest	21472.48	Youth	Overweigh
18	42	female	26.600	0	yes	northwest	21348.71	Adult	Overweigh
19	42	male	24.605	2	yes	northeast	21259.38	Adult	Norma
20	48	male	24.420	0	yes	southeast	21223.68	Adult	Norma
21	53	male	20.900	0	yes	southeast	21195.82	Adult	Norma
22	45	male	22.895	2	yes	northwest	21098.55	Adult	Norma
23	39	male	28.300	1	yes	southwest	21082.16	Adult	Overweigh
24	35	male	27.740	2	yes	northeast	20984.09	Youth	Overweigh
25	36	male	28.025	1	yes	northeast	20773.63	Adult	Overweigh
26	37	female	25.555	1	yes	northeast	20296.86	Adult	Overweigh
27	35	female	28.025	0	yes	northwest	20234.85	Youth	Overweigh
28	39	male	26.410	0	yes	northeast	20149.32	Adult	Overweigh
29	34	male	27.835	1	yes	northwest	20009.63	Youth	Overweigh
30	42	female	23.370	0	yes	northeast	19964.75	Adult	Norma
31	43	female	20.045	2	yes	northeast	19798.05	Adult	Norma
32	32	male	28.930	1	yes	southeast	19719.69	Youth	Overweigh
33	44	female	20.235	1	yes	northeast	19594.81	Adult	Norma

34	37	female	26.400	0	yes	southeast	19539.24	Adult	Overweigh
35	42	male	24.640	0	yes	southeast	19515.54	Adult	Norma
36	40	female	22.220	2	yes	southeast	19444.27	Adult	Norma
37	35	male	24.420	3	yes	southeast	19362.00	Youth	Norma
38	31	male	29.810	0	yes	southeast	19350.37	Youth	Overweigh
39	31	male	25.900	3	yes	southwest	19199.94	Youth	Overweigh
40	33	male	27.100	1	yes	southwest	19040.88	Youth	Overweigh
41	39	female	18.300	5	yes	southwest	19023.26	Adult	Underweigh
42	34	male	25.300	2	yes	southeast	18972.49	Youth	Overweigh
43	43	male	20.130	2	yes	southeast	18767.74	Adult	Norma
44	36	female	22.600	2	yes	southwest	18608.26	Adult	Norma
45	33	male	24.795	0	yes	northeast	17904.53	Youth	Norma
46	32	female	24.600	0	yes	southwest	17496.31	Youth	Norma
47	40	male	19.800	1	yes	southeast	17179.52	Adult	Norma
48	33	female	19.095	2	yes	northeast	16776.30	Youth	Norma
49	38	male	19.300	0	yes	southwest	15820.70	Adult	Norma

```
h_cleaned_processed1 %>%
  group_by(region) %>%
  summarise(
    Min_Charges = min(charges, na.rm = TRUE),
    Max_Charges = max(charges, na.rm = TRUE)
  )
```

```
A tibble: 4 × 3
  region    Min_Charges Max_Charges
  <chr>         <dbl>         <dbl>
1 northeast     1695.         23401.
2 northwest     1621.         22494.
3 southeast     1122.         23563.
4 southwest     1242.         23568.
```

DATA VISUALIZATION

```
summary(h_cleaned_processed1)
```

age		sex		bmi		children	
Min.	:18.00	Length	:1093	Min.	:15.96	Min.	:0.000
1st Qu.:	26.00	N.unique	: 2	1st Qu.:	25.75	1st Qu.:	0.000
Median	:38.00	N.blank	: 0	Median	:29.70	Median	:1.000
Mean	:38.49	Min.nchar:	4	Mean	:29.97	Mean	:1.074
3rd Qu.:	50.00	Max.nchar:	6	3rd Qu.:	33.88	3rd Qu.:	2.000
Max.	:64.00			Max.	:45.43	Max.	:5.000

smoker		region		charges		age_group	
Length	:1093	Length	:1093	Min.	: 1122	Length	:1093
N.unique	: 2	N.unique	: 4	1st Qu.:	4185	N.unique	: 4
N.blank	: 0	N.blank	: 0	Median	: 7731	N.blank	: 0
Min.nchar:	2	Min.nchar:	9	Mean	: 8626	Min.nchar:	3
Max.nchar:	3	Max.nchar:	9	3rd Qu.:	11931	Max.nchar:	5
				Max.	:23568		

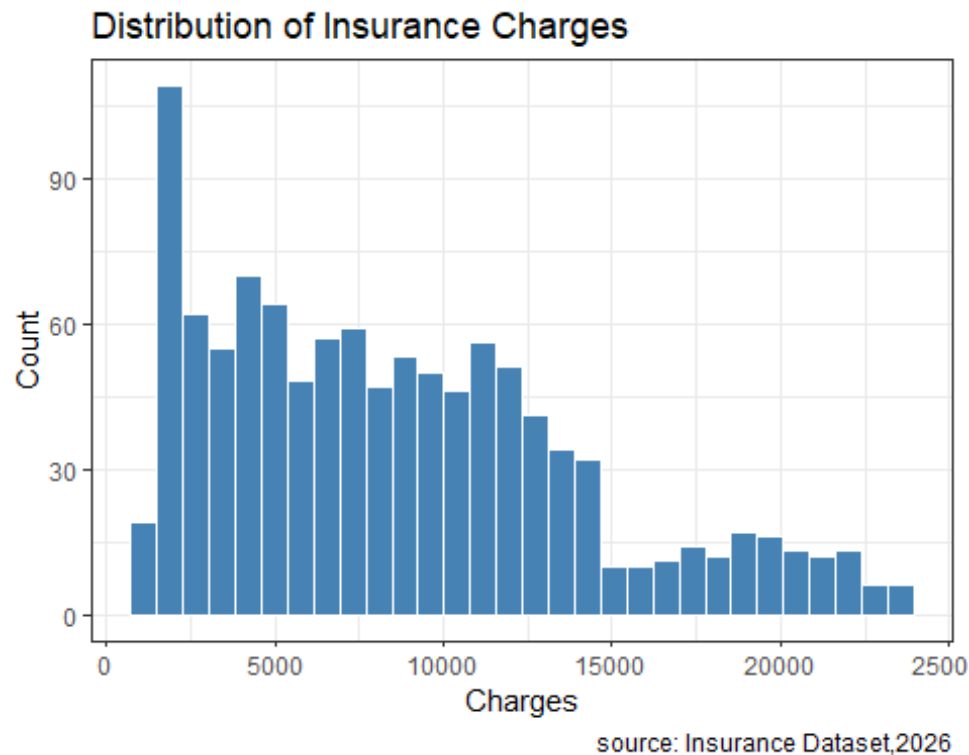
bmi_category	
Length	:1093
N.unique	: 4
N.blank	: 0
Min.nchar:	5
Max.nchar:	11

The data set contains 1,104 observations on demographic characteristics, lifestyle factors, and insurance charges. Respondents range from 18 to 64 years, with an average age of 38 years. The gender distribution is relatively balanced, and non-smokers form the majority of the sample. Insurance charges vary widely, ranging from 1,122 to 23,568, indicating substantial differences in healthcare costs among individuals.

Histogram

```
library(ggplot2)

ggplot(h_cleaned_processed1, aes(x = charges)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Insurance Charges", x = "Charges", y = "Count",
        caption = "source: Insurance Dataset,2026")+
  theme_bw()
```



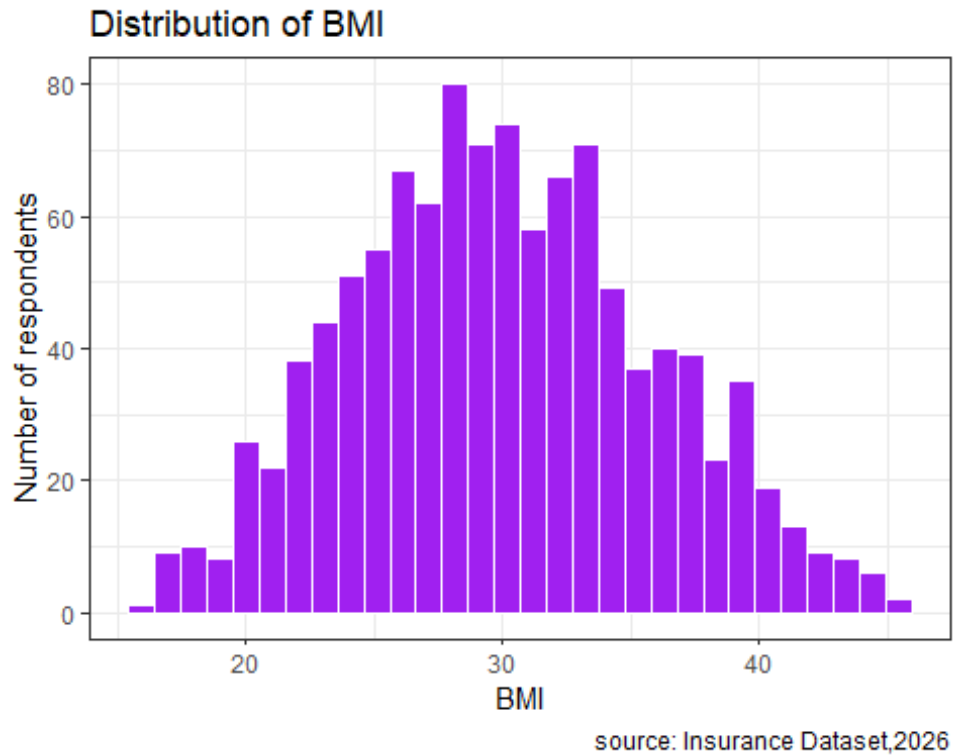
Report of the histogram above

The distribution of insurance charges is right-skewed, with most charges concentrated at lower values and a long tail extending toward higher charges. This suggests that while the majority of individuals incur moderate healthcare expenses, a smaller proportion experience substantially higher medical costs. There also appears to be a small secondary concentration of charges at higher values, which may indicate the influence of specific risk factors such as smoking or obesity.

```
library(ggplot2)

ggplot(h_cleaned_processed1, aes(x = bmi)) +
  geom_histogram(bins = 30, fill = "purple", color = "white") +
  labs(title = "Distribution of BMI", x = "BMI", y = "Number of respondents",
        caption = "source: Insurance Dataset,2026")+

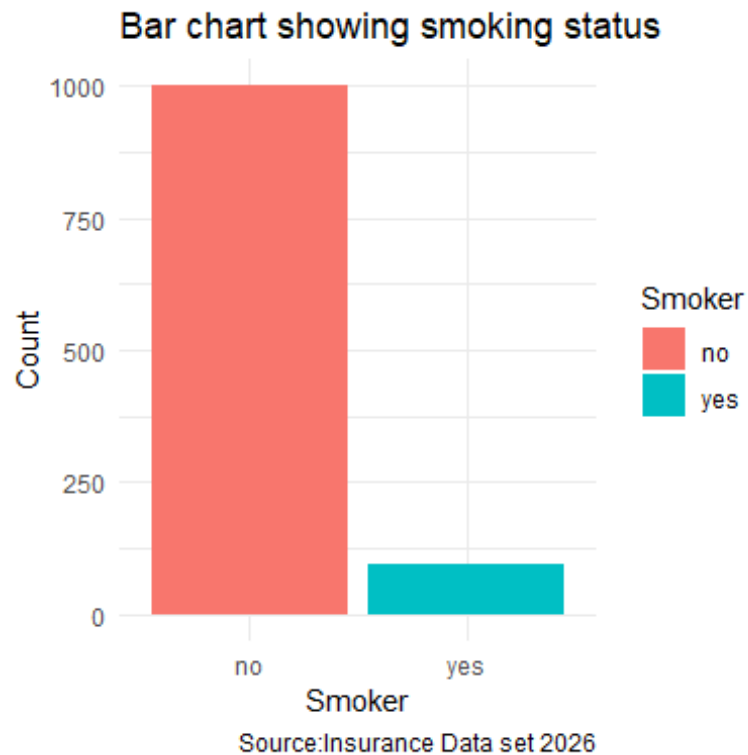
  theme_bw()
```



The histogram shows the distribution of BMI among respondents in the insurance dataset. Most respondents have BMI values between 25 and 35, with the highest concentration around 30. This indicates that many individuals fall within the overweight category. Very low and very high BMI values are less common, showing that extreme BMI levels are rare in the dataset. Overall, the distribution appears approximately normal with moderate variation in BMI values.

```
library(ggplot2)

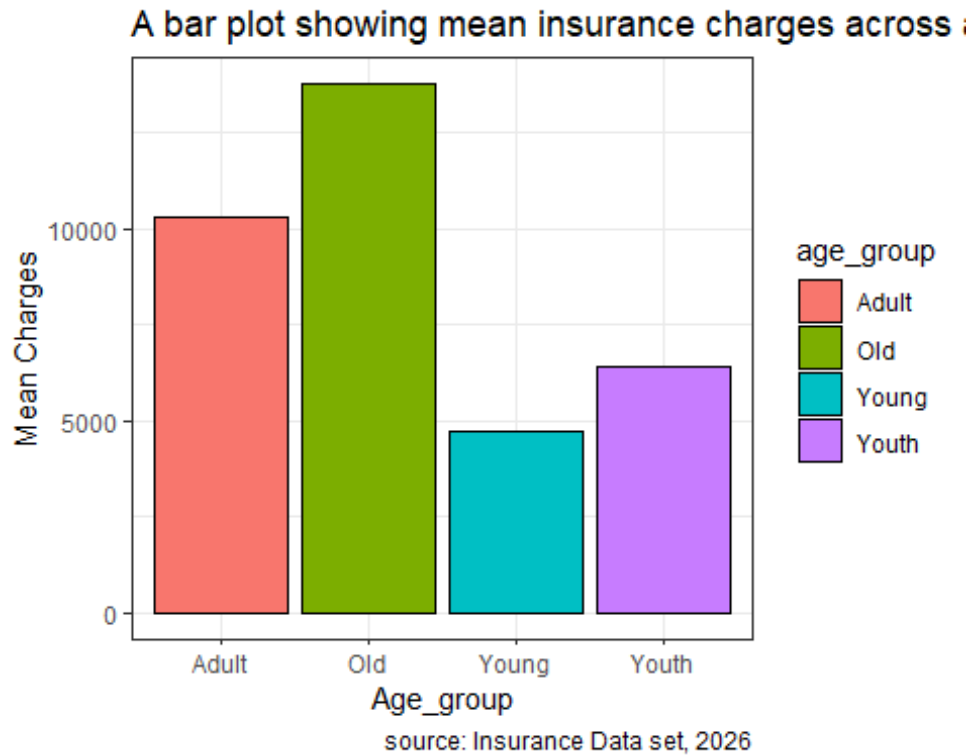
ggplot(h_cleaned_processed1, aes(x = smoker, fill = smoker)) +
  geom_bar() +
  labs(title = "Bar chart showing smoking status", x = "Smoker", y = "Count",
    fill = "Smoker",
    caption = "Source:Insurance Data set 2026") +
  theme_minimal()
```



Report of smokers

Non-smokers make the largest number in the data set

```
h_cleaned_processed1 %>%  
  group_by(age_group) %>%  
  summarise(mean_charges = mean(charges)) %>%  
  ggplot(aes(x = age_group, y = mean_charges, fill = age_group)) +  
  geom_bar(stat = "identity", color = "black") +  
  labs(x = "Age_group",  
       y = "Mean Charges",  
       title = "A bar plot showing mean insurance charges across age groups",  
       caption = "source: Insurance Data set, 2026") +  
  theme_bw()
```



report of the bar plot

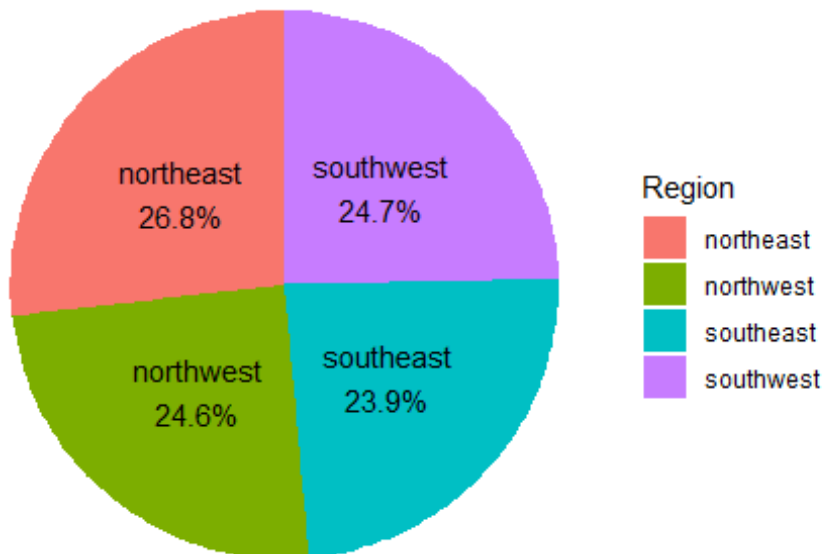
From the results old consumes more insurance(ages over 60) followed by adults(ages between 35 and 59),the young consumes less insurance

pie chart of charges per region

```
library(ggplot2)
library(dplyr)

h_cleaned_processed1 |>
  group_by(region) |>
  summarise(total_charges = sum(charges)) |>
  mutate(pct = total_charges / sum(total_charges),
         label = paste0(region, "\n", scales::percent(pct, accuracy = 0.1)))
|>
  ggplot(aes(x = "", y = total_charges, fill = region)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  geom_text(aes(label = label), position = position_stack(vjust = 0.5), size
= 4) +
  labs(title = "Total Charges by Region", fill = "Region",caption = "Source:I
nsurance Data set,2026"
) +
  theme_void()
```

Total Charges by Region



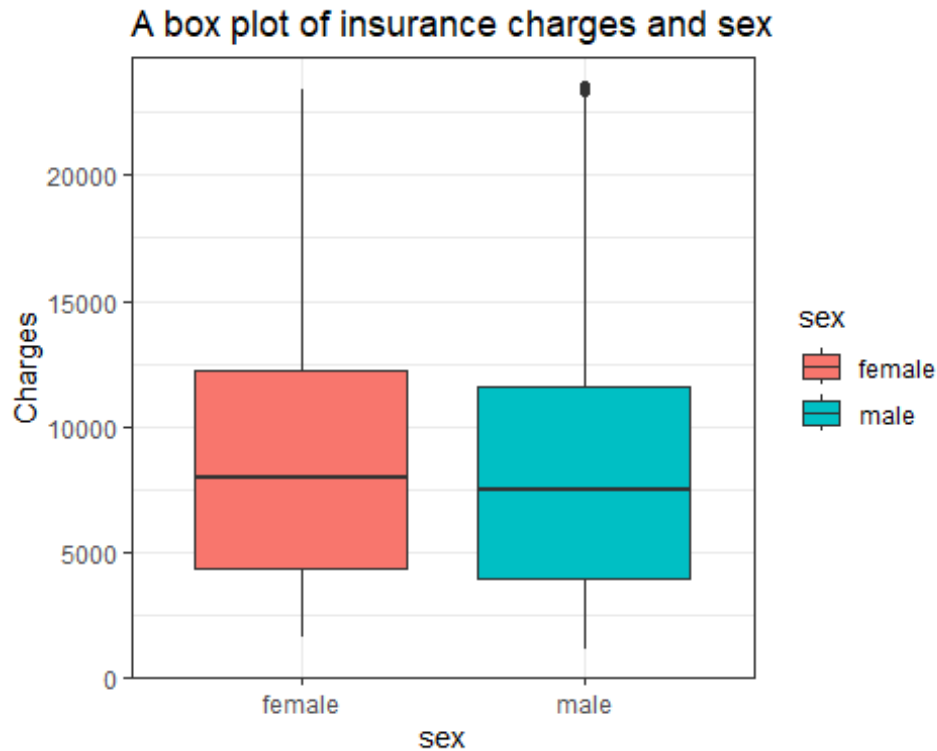
Source: Insurance Data set, 2026

report of the

pie chart The pie chart shows the northeast region accounts for the largest share of total charges at 26.8%, while the other three regions (northwest 24.6%, southwest 24.7%, southeast 23.9%) are fairly evenly split

box plot of sex and charges

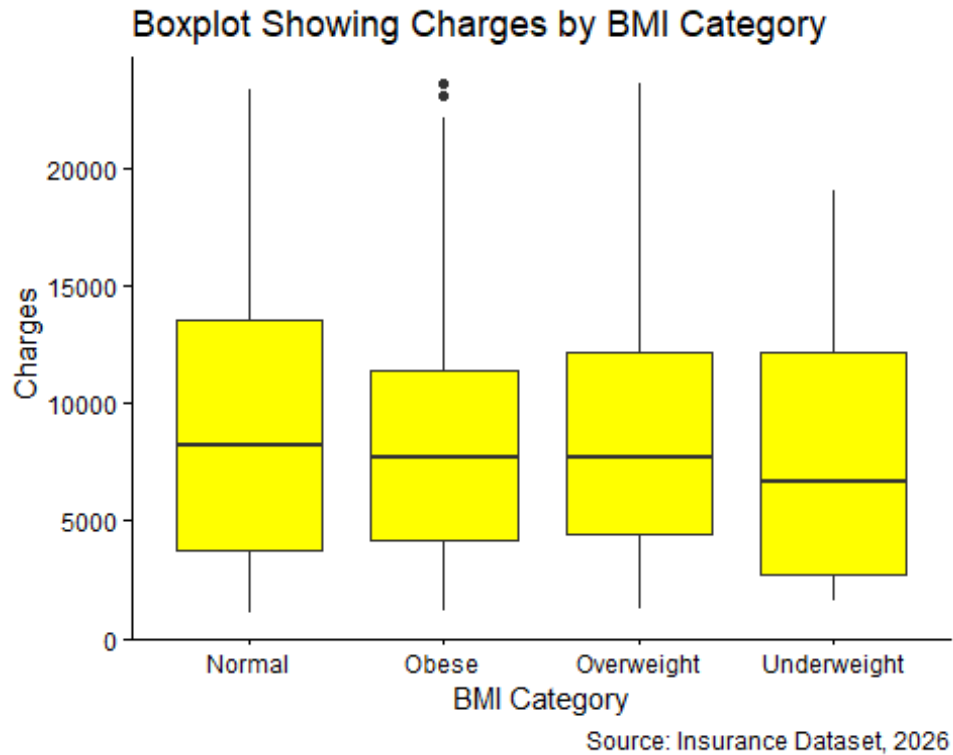
```
h_cleaned_processed1 |>
  ggplot(aes(x=sex, y=charges, fill = sex)) +
  geom_boxplot() +
  labs(x="sex",
       y="Charges",
       title = "A box plot of insurance charges and sex",
       ) +
  theme_bw()
```



Report of insurance charges by sex

The boxplot compares insurance charges between male and female policyholders. Both groups display similar median charges, suggesting that sex on its own does not result in a large difference in typical insurance costs. However, the male group shows a slightly wider interquartile range, indicating greater variability in charges among males. Both groups contain high-end outliers, reflecting that a small number of individuals in each sex incur significantly higher medical costs. Overall, sex appears to have a limited effect on insurance charges in this dataset, and other variables such as smoking status or BMI are likely stronger drivers of cost.

```
h_cleaned_processed1 |>
  ggplot(aes(x = bmi_category, y = charges)) +
  geom_boxplot(fill = "yellow") +
  labs(
    x = "BMI Category",
    y = "Charges",
    title = "Boxplot Showing Charges by BMI Category",
    caption = "Source: Insurance Dataset, 2026"
  ) +
  theme_classic()
```

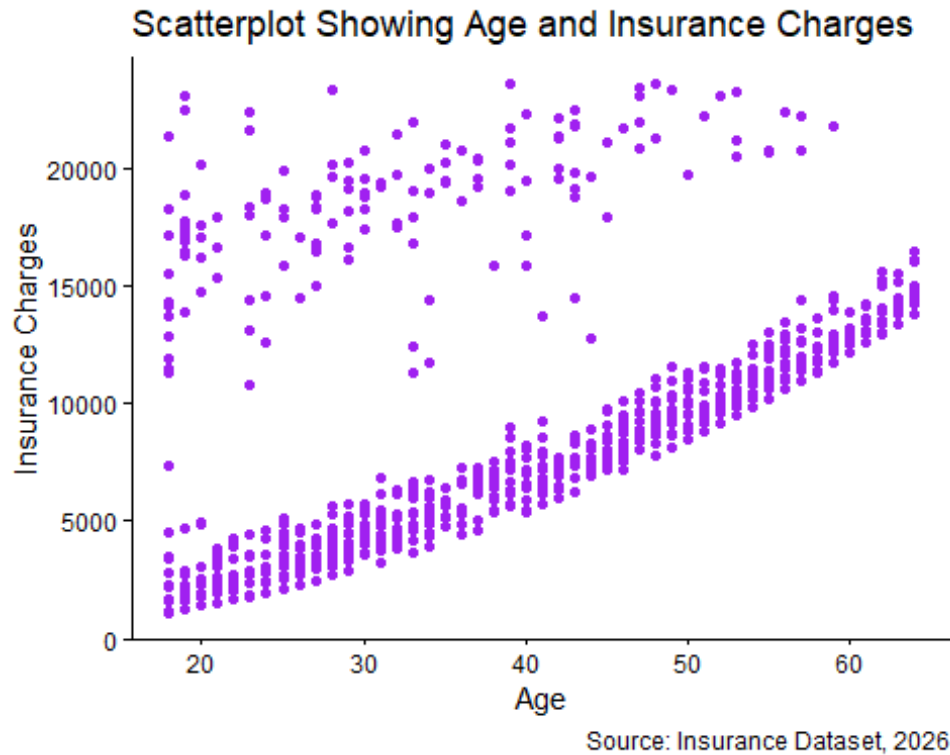


report

The boxplot shows that insurance charges are fairly consistent across BMI categories. Normal weight individuals have the highest median charges, followed closely by the Overweight and Obese groups, which are nearly equal in their central values. Underweight individuals tend to have the lowest charges. The wide spread across all categories suggests that BMI category alone is not a strong predictor of insurance costs.

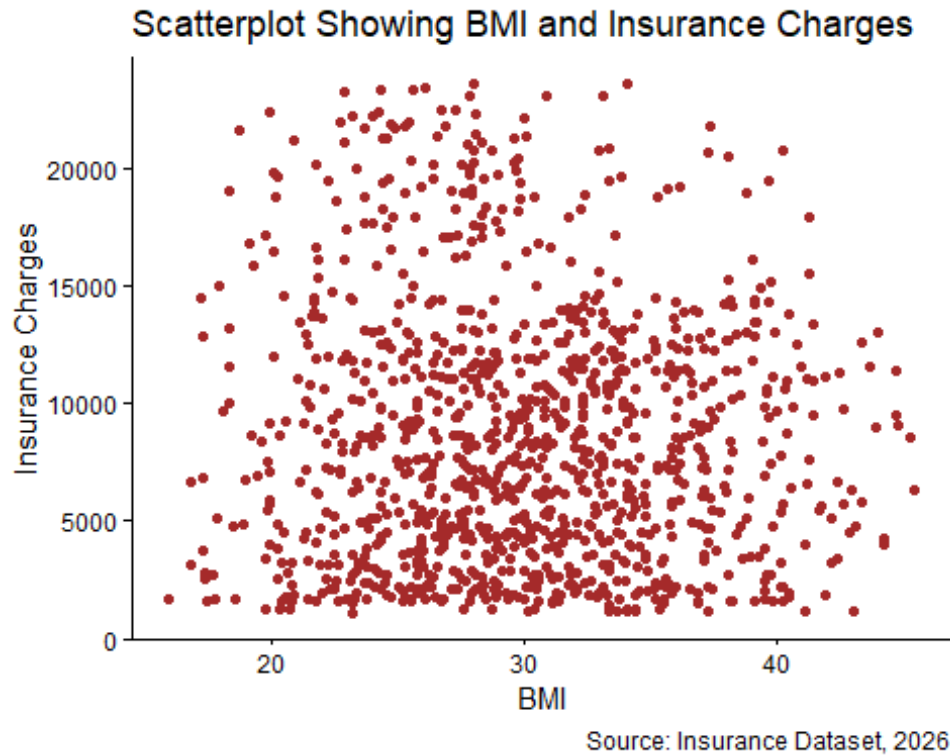
scatter plot of age and insurance

```
h_cleaned_processed1 |>
  ggplot(aes(x = age, y = charges)) +
  geom_point(color = "purple") +
  labs(
    x = "Age",
    y = "Insurance Charges",
    title = "Scatterplot Showing Age and Insurance Charges",
    caption = "Source: Insurance Dataset, 2026"
  ) +
  theme_classic()
```



The scatterplot reveals a positive relationship between age and insurance charges — as age increases, charges tend to rise. Two distinct bands are visible: a lower band where charges increase gradually with age, and an upper band of consistently higher charges across all age groups, likely driven by high-risk individuals such as smokers. This pattern suggests that while age is a contributing factor, other characteristics significantly influence the level of charges.

```
h_cleaned_processed1 |>
  ggplot(aes(x = bmi, y = charges)) +
  geom_point(color = "brown") +
  labs(
    x = "BMI",
    y = "Insurance Charges",
    title = "Scatterplot Showing BMI and Insurance Charges",
    caption = "Source: Insurance Dataset, 2026"
  ) +
  theme_classic()
```



The scatterplot shows no clear linear relationship between BMI and insurance charges. Points are widely scattered across all BMI values, suggesting that BMI on its own is a weak predictor of charges.

heat map

```
library(corrplot)
```

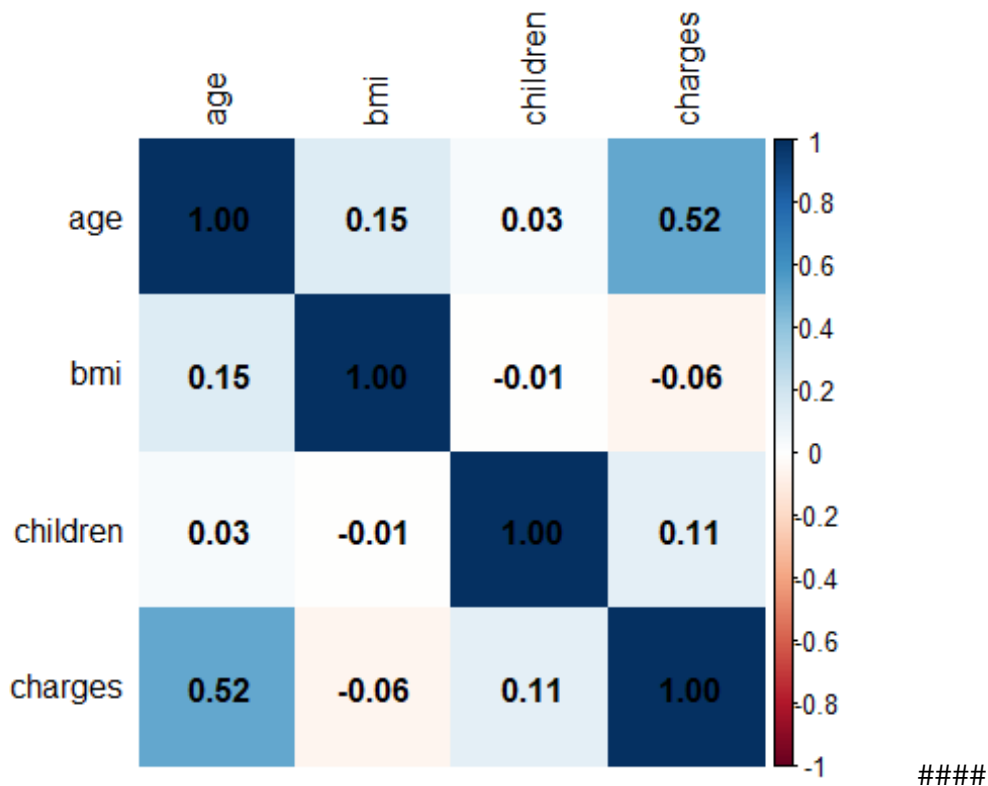
```
corrplot 0.95 loaded
```

```
h_cleaned_processed1 |>
```

```
  select(age, bmi, children, charges) |>
```

```
  cor(use = "complete.obs") |>
```

```
  corrplot(method = "color", addCoef.col = "black", tl.col = "black")
```



The correlation heatmap shows that age has the strongest relationship with insurance charges ($r = 0.52$), while BMI and number of children have negligible correlations with charges. Age is the only numeric variable with a meaningful linear association with insurance costs. Therefore, age should be considered as a key predictor when modeling insurance charges.

Hypothesis testing

t-test #H₀: Mean charges for males and females are equal. #H₁: Mean charges for males and females are different.

T-test

```
t_test<-t.test(charges ~ sex, data = h_cleaned_processed1)
t_test
```

Welch Two Sample t-test

```
data: charges by sex
t = 0.96074, df = 1081.3, p-value = 0.3369
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -330.6017  964.9439
sample estimates:
```

mean in group female	mean in group male
8781.559	8464.388

conclusion

we reject the null hypothesis and conclude that the mean charges between sex is not equal to the mean charges of the other sex.

Chi-square test

H_0 : Smoking and gender are independent. # H_1 : Smoking and gender are associated.

```
table1 <- table(h_cleaned_processed1$sex,  
                h_cleaned_processed1$smoker)
```

```
chisq.test(table1)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table1  
X-squared = 0.1686, df = 1, p-value = 0.6814
```

conclusion

Since the p-value = 0.5739 more than the significance level of 0.05, we fail to reject the null hypothesis (H_0) and conclude that smoking and gender are independent

ho:There is a significant association between gender and smoking status. #h01: There is no significant association between gender and smoking status #ho:There is a significant association between gender and smoking status. #ho2:There is no significant association between gender and smoking status #ho:There is significant association between region and smoking status. #ho3:There is no significant association between region and smoking status.

```
library(gtsummary)
```

```
h_cleaned_processed1 |>  
  select(age, bmi, children, smoker, sex, region, charges) |>  
  tbl_summary(by = smoker) |>  
  add_ci() |>  
  add_p(test = list(  
    all_categorical() ~ "chisq.test",  
    all_continuous() ~ "anova"  
  ))
```

Characteristic	no N = 1,000 ¹	95% CI	yes N = 93 ¹	95% CI	p-value ²
age	39 (26, 51)	38, 40	32 (24, 42)	30, 35	
bmi	30.3 (26.2, 34.3)	30, 31	25.3 (22.6, 27.9)	24, 26	
children					>0.9
0	442 (44%)	41%, 47%	39 (42%)	32%, 53%	
1	243 (24%)	22%, 27%	25 (27%)	18%, 37%	
2	166 (17%)	14%, 19%	16 (17%)	10%, 27%	
3	113 (11%)	9.4%, 13%	10 (11%)	5.6%, 19%	
4	20 (2.0%)	1.3%, 3.1%	2 (2.2%)	0.37%, 8.3%	
5	16 (1.6%)	0.95%, 2.6%	1 (1.1%)	0.06%, 6.7%	
sex					0.7
female	512 (51%)	48%, 54%	45 (48%)	38%, 59%	
male	488 (49%)	46%, 52%	48 (52%)	41%, 62%	
region					0.6
northeast	244 (24%)	22%, 27%	28 (30%)	21%, 41%	
northwest	251 (25%)	22%, 28%	21 (23%)	15%, 33%	
southeast	246 (25%)	22%, 27%	24 (26%)	18%, 36%	
southwest	259 (26%)	23%, 29%	20 (22%)	14%, 31%	
charges	7,153 (3,877, 10,971)	7,391, 7,964	18,972 (17,081, 20,984)	18,305, 19,347	

¹Median (Q1, Q3); n (%)

²Pearson's Chi-squared test

Abbreviation: CI = Confidence Interval

The chi-square test results indicate no significant association between smoking status and sex ($p = 0.6$), region ($p = 0.5$), or number of children ($p > 0.9$), since all p-values are greater than 0.05. However, smokers recorded substantially higher insurance charges compared to non-smokers.

regression analysis

H0: The predictors do not significantly influence insurance charges. #H1: At least one predictor significantly influences insurance charges

```
model_charges <- lm(charges ~ bmi+ age+children+smoker, data = h_cleaned_processed1)
summary(model_charges)
```

Call:

```
lm(formula = charges ~ bmi + age + children + smoker, data = h_cleaned_processed1)
```

Residuals:

Min	1Q	Median	3Q	Max
-3715.4	-1242.8	-738.2	-3.3	20491.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2759.508	545.495	-5.059	4.95e-07	***
bmi	26.653	16.281	1.637	0.102	
age	235.195	6.674	35.243	< 2e-16	***
children	410.977	74.905	5.487	5.09e-08	***
smokeryes	12843.782	339.984	37.778	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3012 on 1088 degrees of freedom

Multiple R-squared: 0.6957, Adjusted R-squared: 0.6946

F-statistic: 621.9 on 4 and 1088 DF, p-value: < 2.2e-16

Conclusion The multiple linear regression results show that age, smoking status, and number of children are significant predictors of insurance charges since their p-values are less than 0.05. In particular, smokers incur substantially higher insurance charges compared to non-smokers. However, BMI is not a statistically significant predictor because its p-value (0.134) exceeds 0.05. The model has an R-squared value of 0.6897, indicating that approximately 68.97% of the variation in insurance charges is explained by the predictors included in the model.